



Gastrodin reversed the traumatic stress-induced depressed-like symptoms in rats

Bombi Lee¹ · Bongjun Sur¹ · Mijung Yeom¹ · Insop Shim^{1,2} · Hyejung Lee¹ · Dae-Hyun Hahn^{1,2}

Received: 27 January 2016 / Accepted: 12 May 2016 / Published online: 14 July 2016
© The Japanese Society of Pharmacognosy and Springer Japan 2016

Abstract Exposure to severe stress can lead to the development of neuropsychiatric disorders such as depression and post-traumatic stress disorder (PTSD) in at-risk individuals. Gastrodin (GAS), a primary constituent of an Oriental herbal medicine, has been shown to effectively treat various mood disorders. Thus, the present study aimed to determine whether GAS would ameliorate stress-associated depression-like behaviors in a rat model of single prolonged stress (SPS)-induced PTSD. Following the SPS procedure, rats received intraperitoneal administration of GAS (20, 50, or 100 mg/kg) once daily for 2 weeks. Subsequently, the rats performed the forced swimming test, and norepinephrine (NE) levels in the hippocampus were measured. Daily GAS (100 mg/kg) significantly reversed depression-like behaviors and restored SPS-induced increases in hippocampal NE concentrations as well as tyrosine hydroxylase expression in the locus coeruleus. Furthermore, the administration of GAS attenuated SPS-induced decreases in the hypothalamic expression of neuropeptide Y and the hippocampal mRNA expression of brain-derived neurotrophic factor. These findings indicate that GAS possesses antidepressant effects in the PTSD and may be an effective herbal preparation for the treatment of PTSD.

Keywords Post-traumatic stress disorder · Single prolonged stress · Depression · Norepinephrine · Gastrodin

Introduction

Post-traumatic stress disorder (PTSD) is a debilitating neuropsychiatric or psychological disorder that may develop in a vulnerable subset of individuals after exposure to a traumatic event [1]. This psychopathological response to traumatic stressors is characterized by intrusive recollections of the traumatic event, avoidance of reminders of the event, hyperarousal, and is considered to be outside the realm of normal human experience [2, 3]. PTSD is often co-morbid with major depression and anxiety as well as substance abuse, sleep disturbances, and marked psychosocial and occupational impairments [4]. The symptoms of PTSD are thought to reflect trauma-induced changes that can lead to long-term dysfunction in the regulation of stress and an inappropriate adaptation of the hypothalamic–pituitary–adrenal (HPA) axis [5, 6]. Therefore, hyperarousal and dysregulation within the HPA axis are closely associated with the development of PTSD and are linked to serious changes in affective behaviors that are indicative of or consistent with depression-like symptoms [7]. Moreover, the dysregulation of noradrenergic neurons has been associated with intrusive recollections that are attributable to PTSD [8, 9]. In the brain, norepinephrine (NE) is synthesized in the locus coeruleus (LC), and elevated levels may modulate the pathophysiology of various mood disorders including PTSD [10, 11].

The first line treatment for PTSD is antidepressants, which have been shown to efficaciously reduce symptom severity and improve relapse prevention in PTSD patients [12]. However, there are several shortcomings associated

✉ Bombi Lee
bombi@khu.ac.kr

¹ Acupuncture and Meridian Science Research Center, College of Korean Medicine, Kyung Hee University, 26 Kyunghedae-ro, Dongdaemun-gu, Seoul 02447, Republic of Korea

² The Graduate School of Basic Science of Korean Medicine, College of Korean Medicine, Kyung Hee University, Seoul 02447, Republic of Korea

with antidepressant treatments including their limited efficacy and undesirable side effects, and; thus, there is a major unmet medical need for novel treatment approaches for patients with PTSD [13, 14].

The neuroendocrine findings specific to PTSD have served as the basis for animal models that are useful for elucidating the pathophysiology of PTSD [15]. Single prolonged stress (SPS) has been found to reproduce the cardinal symptoms of PTSD [4]. SPS causing HPA axis dysregulation may lead to profound maladaptive changes manifested in behaviors that resemble depression, thereby mimicking human mental disorders involving these states [1].

Gastrodia elata Blume (GE) is one of the most popular traditional medicinal herbs used in East and Southeast Asian countries [16]. Gastrodin (GAS), a main bioactive component of GE, has been widely used in clinical populations for the treatment of neurasthenia, dizziness, epilepsy, migraine, and dementia [17]. GAS penetrates the blood-brain barrier and enters the central nervous system [18], and a considerable amount of evidence has demonstrated that this pharmacotherapy may have neuroprotective effects [19–21]. Recent findings suggest that GAS is neuroprotective against hypoxia in cultured cortical neurons, protects primary cultured rat hippocampal neurons against amyloid- β -peptide-induced neurotoxicity [22], and ameliorates cerebral damage following transient focal cerebral ischemia [23]. Additionally, GAS inhibits the expressions of inducible nitric oxide (NO) synthase, cyclooxygenase-2, and pro-inflammatory cytokines in cultured lipopolysaccharide (LPS)-stimulated microglia via mitogen-activated protein kinase (MAPK) pathways [20]. These actions may improve learning capabilities and facilitate memory consolidation and retrieval [12, 24]. GAS also exerts anxiolytic-like effects via the GABAergic system [25] and ameliorates depression-like behaviors in rats exposed to chronic unpredictable stress [16]. Thus, the present study hypothesized that GAS would be protective against the development of depression-like behaviors, which represents the core symptoms of PTSD-related abnormalities in a rat model of PTSD using single prolonged stress (SPS). We measured the forced swimming test (FST) as a behavioral measure and brain concentration of NE in the hippocampus and tyrosine hydroxylase (TH) expression in the hypothalamus as a neurobiological measure of GAS action and potential underlying mechanisms.

Materials and methods

Animals and GAS administration

The present study utilized adult male Sprague–Dawley (SD) rats weighing 200–220 g (6 week-old, Samtako Animal

Company; Seoul, Korea). All rats were housed in a limited access rodent facility with five rats per polycarbonate cage. The room controls were set to maintain the temperature at 22 ± 2 °C and the relative humidity at 55 ± 15 %, the cages were lit by artificial light during 12-h light/dark cycle (light 07:00–19:00) each day, and sterilized drinking water and a standard chow diet were supplied ad libitum throughout the experiments. All animal experiments began a minimum of 7 days after the animals arrived, were conducted in accordance with the *Guide for the Care and Use of Laboratory Animals* from the National Institutes of Health (NIH Publications No. 80–23; revised in 1996), and were approved by the Kyung Hee University Institutional Animal Care and Use Committee. All efforts were made to minimize the number and suffering of animals.

Different groups of rats, 6–7 animals per group, were used for drugs treatment and tests. The rats were randomly divided into six groups of six to seven individuals each as follows: normal group (0.9 % physiological saline (SAL)-treated group, $n = 6$), the SPS-induced plus SAL-treated group (SPS group as a control, $n = 7$), the SPS-induced plus 20 mg/kg GAS-treated group (SPS + GAS20 group, $n = 7$), the SPS-induced plus 50 mg/kg GAS-treated group (SPS + GAS50 group, $n = 7$), the SPS-induced plus 100 mg/kg GAS-treated group (SPS + GAS100 group, $n = 7$), and the SPS-induced plus 10 mg/kg fluoxetine-treated group (SPS + FLX group, $n = 6$). GAS (20, 50, and 100 mg/kg body weight; Sigma-Aldrich Chemical Co., St. Louis, MO, USA) and the positive drug FLX (10 mg/kg, fluoxetine hydrochloride; Sigma-Aldrich Chemical Co. St. Louis, MO, USA) were administered intraperitoneally (i.p.) in a volume of 1 ml/kg for 14 days after the SPS procedure. The GAS and FLX were dissolved in SAL before use.

Single prolonged stress (SPS)

The SPS model of PTSD was performed as previously described [26]. Briefly, rats were first immobilized for 2 h on restraint tubes (20 cm height, 7 cm diameter). Following immobilization, rats were immediately subjected to forced swim for 20 min in a Plexiglass cylinder (50 cm height, 20 cm diameter) filled to two-thirds full with 24 °C fresh water. The animals were dried and allowed to recuperate for 15 min and then exposed to ether vapor until loss of consciousness. The SPS procedure was performed between 10 AM and 4 PM. Following SPS stressors, animals were housed one per cage and left undisturbed for 14 days to allow PTSD-like symptoms to manifest [26]. The following parameters were measured to monitor the effects of the development of psychiatric disorders by the SPS model of PTSD: changes of body weight gains and serum corticosterone (CORT) levels. Behavioral testing for depression-like behaviors was done 24 h after the end of the undisturbed

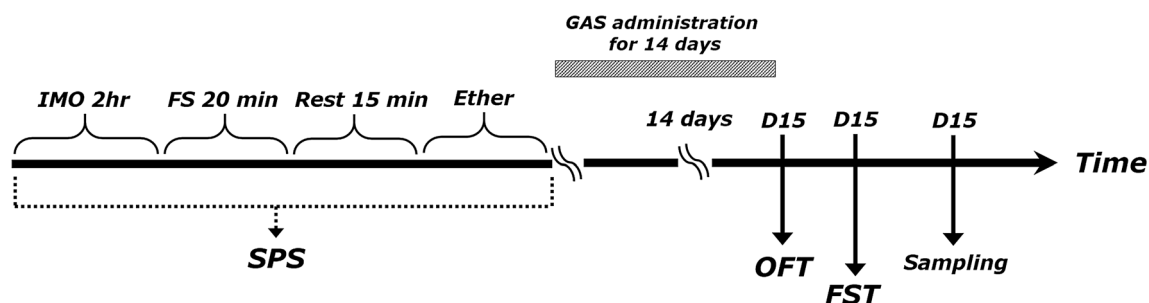


Fig. 1 Experimental schedules for the SPS-induced depression-like behaviors and GAS treatment in the rats. Different groups of rats ($n = 6-7$) were used for all experiments

condition. The entire experimental schedules of SPS, behavioral examinations, and drug administration are shown in Fig. 1. To avoid carryover from one test to another, each behavioral test was performed separately or in an interval between each behavioral test. Following behavioral testing and the measurement of body weight, all rats were sacrificed and their brain tissues were collected immediately for use in the study or stored at -70°C for later use.

Forced swimming test (FST)

The FST used was similar to that described in detail elsewhere [27], with some modification. Briefly, the test was done by placing a rat in a transparent Plexiglas cylinder (20 cm diameter $\times$ 50 cm height), which was filled up to a depth of 30 cm with water at 25°C . At this depth, rats could not touch the bottom of the cylinder with their tails or hind limbs. Two swimming sessions were conducted: an initial 15 min pretest, followed by a 5 min test 24 h later. Animals were subjected to 5 min of forced swim, and escape behaviors (climbing and swimming) were determined. The animals' behavior was continuously recorded by experimenter-manual scoring during the testing session with an overhead video camera to tape behavior for later manual scoring. Immobility behavior was calculated as the length of time in which the animal did not show escape responses (e.g., total time of the test minus time spent in climbing and swimming behaviors). The rats were judged to be immobile when they remained in the water without struggling and were making only those movements necessary to keep its head above water. Climbing behavior was defined as upward-directed movements of the forepaws along the side of the swim chamber and swimming behavior was considered as movements throughout the swim chamber including crossing into another quadrant.

CORT and norepinephrine (NE) analyses

After PTSD for 14 days, CORT concentration in blood, and NE concentration in brain tissue were determined. Animals

were killed by decapitation one day after behavioral measurement. For this, the unanesthetized rats were rapidly decapitated, perfused through the ascending aorta with SAL, and blood was quickly collected via the abdominal aorta. The hippocampus was rapidly removed from the rat brains in randomized order. Special care was taken to avoid pre-decapitation stress; while rats were rapidly decapitated, the other animals were left outside the room and handled for a few minutes prior to sampling. The blood samples were centrifuged at 4000 g for 10 min, and serum was collected and stored at -20°C until use. The CORT concentration was measured by a competitive enzyme-linked immunoassay (ELISA) using a rabbit polyclonal CORT antibody (Corticosterone ELISA kit; Novus Biologicals, LLC., Littleton, CO, USA) according to the manufacturer's protocol. The brain tissue samples were stored at -80°C until use. Hippocampus was homogenized in a lysis buffer containing 137 mM NaCl, 20 mM Tris (pH 8.0), 1 % NP40, 10 % glycerol, 1 mM PMSF, 10 mg/ml aprotinin, 1 mg/ml leupeptin, and 0.5 mM sodium vanadate. Homogenization was carried out on ice using a tissue homogenizer and incubated for 1 min at 4°C with shaking. Homogenates were centrifuged and supernatants were collected. Protein concentrations were estimated by the procedure of Gmeiner et al. [28] with BSA as the standard. The NE concentration was measured by a competitive enzyme-linked immunoassay (ELISA) using a mouse monoclonal NE antibody (NE ELISA kit; Novus Biologicals, LLC., Littleton, CO, USA) according to the manufacturer's protocol. After proper color developed, the optical density was measured at 450 nm using an ELISA reader (MutiRead 400; Authos Co., Vienna, Austria).

Immunohistochemical analyses of tyrosine hydroxylase (TH) and neuropeptide Y (NPY)

To conduct the immunohistochemical analyses, three rats from each group were deeply anesthetized with sodium pentobarbital (80 mg/kg, i.p.) and perfused through the ascending aorta with SAL followed by 300 ml 4 %

paraformaldehyde in 0.1 M phosphate-buffered saline (PBS). The brains were removed in a randomized order, post-fixed overnight, and cryoprotected with 20 % sucrose in 0.1 M PBS at 4 °C. Coronal Sections (30 µm) were cut through the LC and hypothalamus using a cryostat (Leica CM1850, Leica Microsystems Ltd.; Nussloch, Germany) according to the rat atlas of Paxinos and Watson [29]. The sections obtained from the rats were immunostained for TH and NPY expression using the avidin–biotin–peroxidase complex (ABC) method. Briefly, the sections were incubated with primary sheep anti-TH antibody (1:2000 dilution; #AB1542, Chemicon International Inc., Temecular, CA, USA) and rabbit anti-NPY antibody (1:2000 dilution, #22940, Immunostar, Hudson, WI, USA) in PBS plus 0.3 % Triton X-100 (PBST) for 72 h at 4 °C. Next, the sections were incubated for 120 min at room temperature with secondary antibodies (1:200 dilution, Vector Laboratories Co.; Burlingame, CA, USA) in PBST containing 2 % normal goat serum. To visualize immunoreactivity, the sections were incubated for 90 min in ABC reagent (Vectastain Elite ABC kit, Vector Labs. Co.), and then in a solution containing 3,3'-diaminobenzidine (DAB; Sigma-Aldrich) and 0.01 % H₂O₂ for 1 min. Finally, the tissues were washed in PBS, briefly rinsed in distilled water, and mounted individually onto slides.

Images were captured using the AxioVision 3.0 imaging system (Carl Zeiss, Inc.; Oberkochen, Germany) and processed using Adobe Photoshop (Adobe Systems, Inc.; San Jose, CA, USA). The sections were viewed at 200× magnification, and the expression of TH in the LC and NPY in the PVN were measured. TH- and NPY-labeled cells were counted by an observer blinded to the experimental groups. The counted sections were randomly chosen from equal levels of serial sections along the rostral-caudal axis. The cells within the LC and hypothalamus areas were counted on each of the five sections per animal. Counting the immunopositive cells was performed within the square (150 × 150 µm²), anatomically localized in LC and hypothalamus sections per rat brain according to the stereotactic rat brain atlas of Paxinos and Watson [29]. Immunohistochemical analysis of tissue samples from three different animals was enough to show the changes of protein markers at molecular levels, as published in our previous studies [29]. Distinct brown spots indicating TH- and NPY-immunopositive cells were observed in the LC and hypothalamus. The TH- and NPY-immunopositive cells were only counted when their densities reached a defined darkness above the background level. The brightness and contrast of all images were not adjusted to remove any subjective selection of immunoreactive cells.

Total RNA preparation and reverse transcription-polymerase chain reaction

The mRNA expression levels of BDNF in hippocampal tissues isolated from four rats per group were determined using reverse transcription-polymerase chain reaction (RT-PCR). Following decapitation, the brains were perfused through the ascending aorta with SAL, and removed quickly and stored at −80 °C until use. Total RNA was prepared from the brain tissue samples using TRIzol[®] reagent (Invitrogen Co.; Carlsbad, CA, USA) according to the supplier's instructions. Briefly, complementary DNA was synthesized from total RNA using reverse transcriptase (Takara Co.; Shiga, Japan). Then, RT-PCR was performed using a PTC-100 programmable thermal controller (MJ Research, Inc.; Watertown, MA, USA) as follows: for glyceraldehyde-3-phosphate dehydrogenase (GAPDH), 30 cycles of denaturation at 95 °C for 30 s, annealing at 58 °C for 30 s, and extension at 72 °C for 30 s; for BDNF, 27 cycles of denaturation at 95 °C for 30 s, annealing at 57 °C for 30 s, and extension at 72 °C for 30 s.

All primers were designed using published mRNA sequences and primer designing software (Primer 3, Whitehead Institute for Biomedical Research; Cambridge, MA, USA; www.genome.wi.mit.edu) with the following sequences: for GAPDH (223 base pairs [bp]), (forward) 5'-ATC CCA TCA CCA TCT TCC AG-3' and (reverse) 5'-CCT GCT TCA CCA CCT TCT TG-3'; for BDNF (153 bp), (forward) 5'-CAG GGG CAT AGA CAA AAG-3' and (reverse) 5'-CTT CCC CTT TTA ATG GTC-3'. The PCR products were separated on 1.2 % agarose gels and stained with ethidium bromide, and the density of each band was quantified using an image-analyzing system (i-MaxTM, CoreBio System Co.; Seoul, Korea). The expression levels of BDNF were compared by calculating the relative density of BDNF, with that of GAPDH.

Statistical analysis

All measurements were performed by an independent investigator blinded to the experimental conditions, and the results are expressed as mean ± standard error of the mean (SE). Differences within or between normally distributed data were analyzed using an analysis of variance (ANOVA) using SPSS (Version 13.0, SPSS, Inc.; Chicago, IL, USA) and a Tukey's post hoc test. A *p* value of <0.05 was considered to indicate statistical significance.

Results

Effect of GAS on changes in body weight and serum CORT levels following single prolonged stress

We measured the body weight of each rat in each group for 14 days. In the present study, rats exposed to SPS began to lose body weight on the first day of the procedure (difference between daily weights and starting weight; Fig. 2a). Compared with the SAL-treated groups, rats in the SPS group showed a significant gradual reduction in body weight gain over 14 days. However, rats treated with 20 or 100 mg/kg GAS exhibited no significant differences of this reduction in body weight gain compared with the SPS group. Also, rats treated with 50 mg/kg GAS showed a linear reduction of initial body weight from day 1 to 4 (day 3: -3.3 ± 4.6 g) and then a linear increase from day 5 to 14 (Fig. 2a). In rats treated with 50 mg/kg GAS, the decrease in body weight gain induced by GAS administration is not related to changes in GAS concentration. Similarly, the reduced body weight in the SPS + GAS50 group was not associated with a reduction in food intake. Subjects treated with FLX exhibited no significant differences of this increase in body weight gain compared with the SAL group.

Serum CORT levels were also measured in each group following the SPS for 14 days (Fig. 2b). An ELISA revealed that SPS significantly increased serum CORT concentration by 245.13 % compared to saline-treated rats ($p < 0.05$). Thus, the SPS procedure induced depression-like symptoms in rats, and this was exploited to develop the PTSD. However, the administration of GAS (50 or 100 mg/kg) significantly inhibited the SPS-induced increase in serum CORT levels ($p < 0.05$).

Effect of GAS on depression-like behavior following single prolonged stress

To examine the effect of GAS on response to traumatic stress, rats were treated with GAS or saline 30 min after SPS. Rats subjected to the SPS and undisturbed situations for 14 days exhibited a significant depression-like phenotype, characterized by increased duration of immobility during the FST compared to saline-treated controls (Fig. 3). Rats in the SPS group exhibited more immobility during the FST compared to the SAL group ($p < 0.01$; Fig. 3a). Long-term treatment with GAS (100 mg/kg) for 14 days displayed a significant decrease in durations of immobility during 5 min in the FST compared to SPS group ($p < 0.05$), indicating that the administration of 100 mg/kg GAS decreased depression-like behaviors. Another key behavior, climbing behavior, was also analyzed. Rats in the SPS group exhibited a significant decrease in climbing behavior during the FST relative to the SAL group ($p < 0.01$; Fig. 3b). Furthermore, compared with the SPS group, rats in the SPS + GAS100 group exhibited slightly a restoration of climbing behavior time during 5 min in the FST, although the findings were not significant. This indicated that GAS administration significantly restored depression-like despair behaviors. However, SPS over 14 days did not induce significant differences in swimming behaviors among the groups during the FST ($p = 0.415$; Fig. 3c). These results reveal that the recovery in immobility during depression-like behaviors in the SPS + GAS100 group were almost comparable to those of the SPS + FLX group.

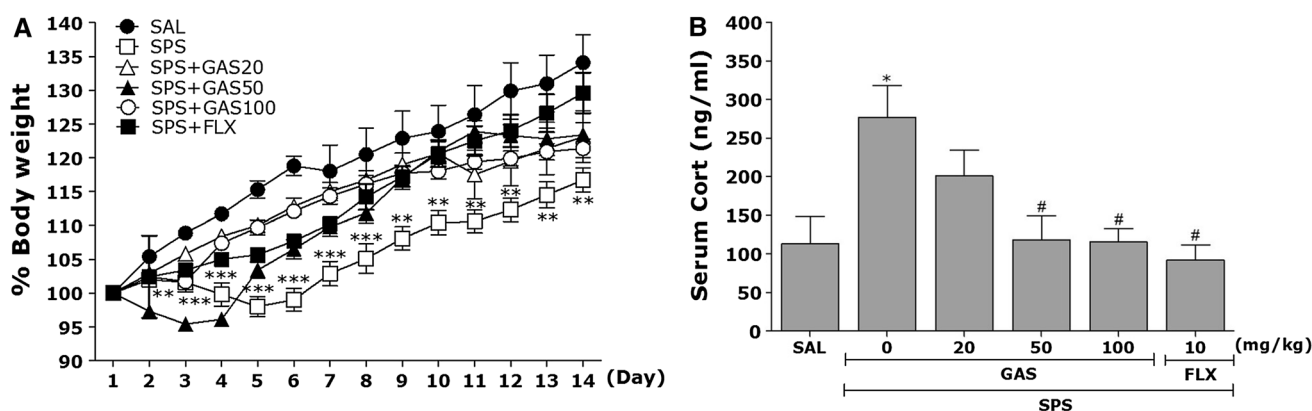


Fig. 2 Effects of GAS administration on body weights gain (a) and CORT serum levels (b) in rats exposed to SPS. * $p < 0.05$, ** $p < 0.01$, *** $p < 0.001$ vs. SAL group; # $p < 0.05$ vs. SPS group

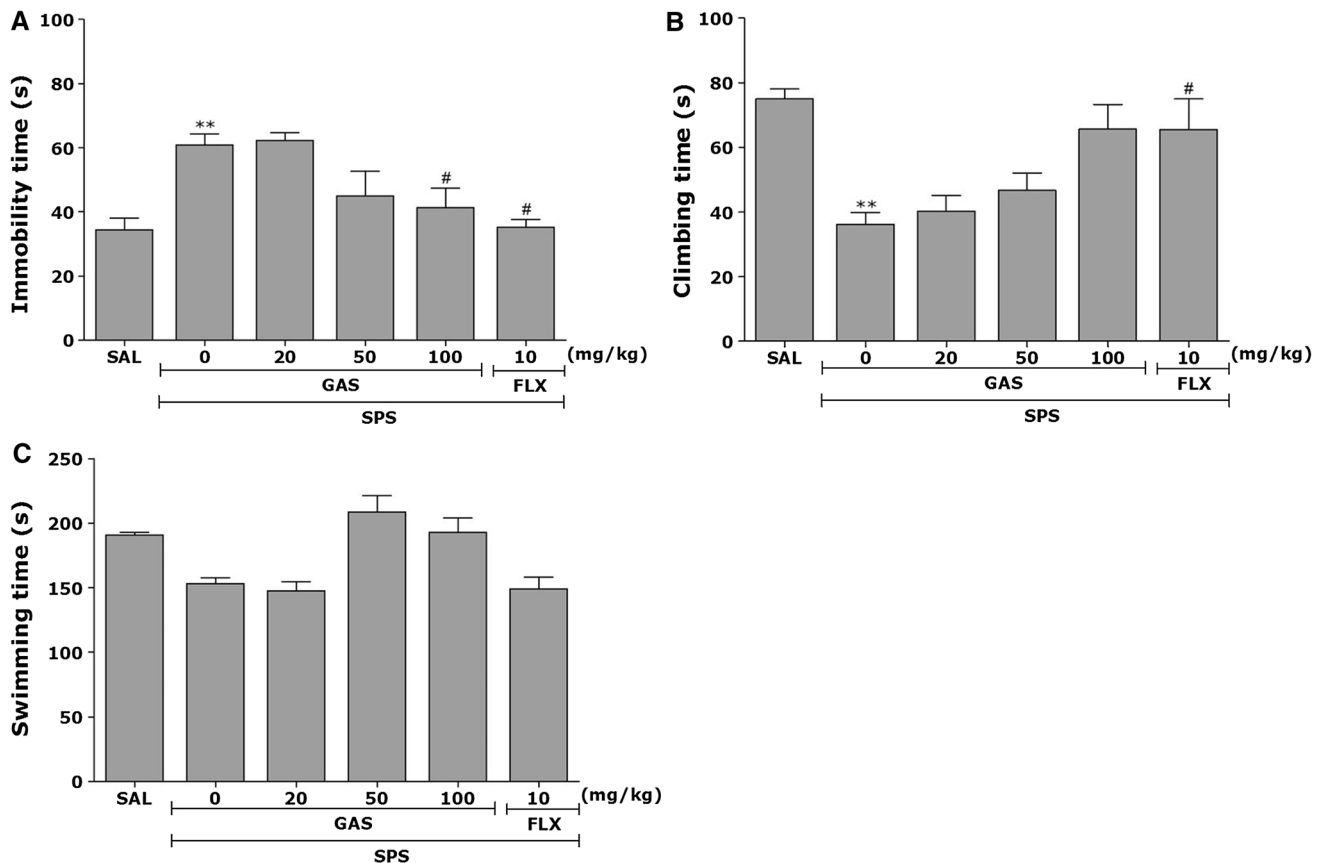


Fig. 3 Effects of GAS administration on immobility time (a), climbing behavior (b), and swimming time (c) in the FST following SPS. ** $p < 0.01$ vs. SAL group; # $p < 0.05$ vs. SPS group

Effect of GAS on norepinephrine concentrations in the hippocampus following single prolonged stress

The ELISA analysis demonstrated that rats exposed to SPS for 14 days significantly increased the NE concentration in the hippocampus by 247.32 %, compared with rats in the non-treated SAL group ($p < 0.05$; Fig. 4). Daily

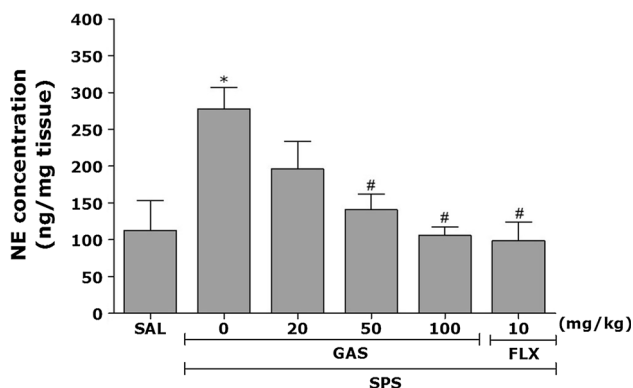


Fig. 4 Effect of GAS administration on NE concentration in the hippocampus of rats exposed to SPS. * $p < 0.05$ vs. SAL group; # $p < 0.05$ vs. SPS group

administration of GAS showed it significantly decreased the SPS-induced increase of NE concentration in the hippocampus, as compared to the SPS group ($p < 0.05$). The results also indicated that the NE protein concentration in the hippocampus in rats receiving 10 mg/kg FLX was similar to that in rats receiving 100 mg/kg GAS.

Effect of GAS on TH- and NPY-like immunoreactivities following single prolonged stress

After the behavioral measures were completed, TH-like immunoreactivity was also analyzed in the central adrenergic regions, including the LC (Fig. 5a and b). In the rat brains in the SPS group, the numbers of TH immunoreactive fibers in the LC were increased to 163.52 %. Analysis of the numbers of TH-immunoreactive neurons values revealed that rats exposed to SPS showed a significant increase in TH expression, as compared to the SAL group ($p < 0.01$). The number of TH-immunoreactive neurons significantly decreased in the central adrenergic regions in the SPS + GAS100 group, as compared to those in the SPS group ($p < 0.05$). This finding suggests that the increased TH-immunoreactivity induced by rats exposed to

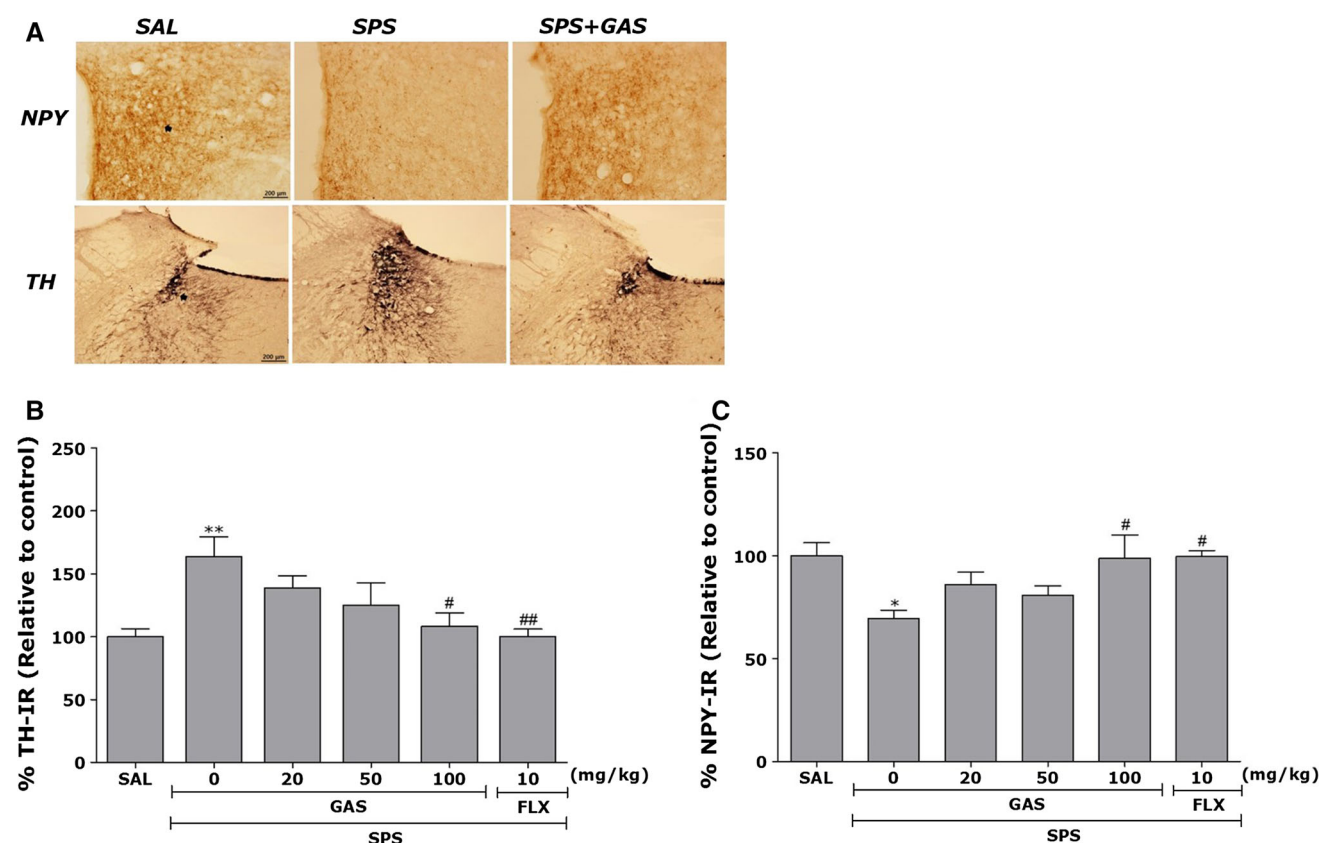


Fig. 5 Effects of GAS administration on the NPY expression levels in the PVN of the hypothalamus and TH expression in the LC. Representative images (a) and the relative percentage values (b and

c) are shown. * $p < 0.05$, ** $p < 0.01$ vs. SAL group; # $p < 0.05$ vs. SPS group

SPS for 14 days was significantly restored by FLX administration ($p < 0.01$).

NPY-like immunoreactivity was analyzed in the cell bodies of various hypothalamic regions including the paraventricular nucleus (PVN; Fig. 5b and c). Compared with the SAL group, NPY-like immunoreactive cells in the PVN of the SPS group decreased to 69.60 %, which indicated that SPS results in a significant decrease in NPY expression. Compared with the SPS group, there was a significant increase in the number of NPY-immunoreactive neurons in the hypothalamus of the SPS + GAS100 group ($p < 0.05$). This finding indicated that the decrease in the number of NPY-immunoreactive neurons in rats exposed to SPS was significantly attenuated by the administration of GAS. This finding suggests that the decreased NPY-immunoreactivity induced by rats exposed to SPS for 14 days was significantly restored by FLX administration ($p < 0.05$).

Effect of GAS on BDNF mRNA in the hippocampus following single prolonged stress

The effect of GAS administration on the expression of BDNF mRNA in the hippocampus of rats exposed to SPS

was investigated using RT-PCR (Fig. 6). BDNF mRNA expression levels were normalized against those of GAPDH mRNA, the internal control. There was a significant decrease in the hippocampus BDNF mRNA expression in the SPS group compared with the SAL group ($p < 0.01$). The decreased expression of BDNF mRNA in the SPS group was significantly restored to levels similar to those in the SAL group by 100 mg/kg GAS ($p < 0.05$). This indicated that the expression of BDNF mRNA in the hippocampus of rats receiving 100 mg/kg GAS was similar to that of rats receiving 10 mg/kg FLX.

Discussion

The present study found that GAS had antidepressant-like effects in rats exposed to the SPS-induced pathogenesis associated with clinical psychological disorders. More specifically, following the SPS procedure, GAS (50 and 100 mg/kg) decreased the duration of immobility in the FST. However, this effect was significant only at a dose of 100 mg/kg, which is consistent with the findings of a previous study [12]. It is likely that these behavioral effects

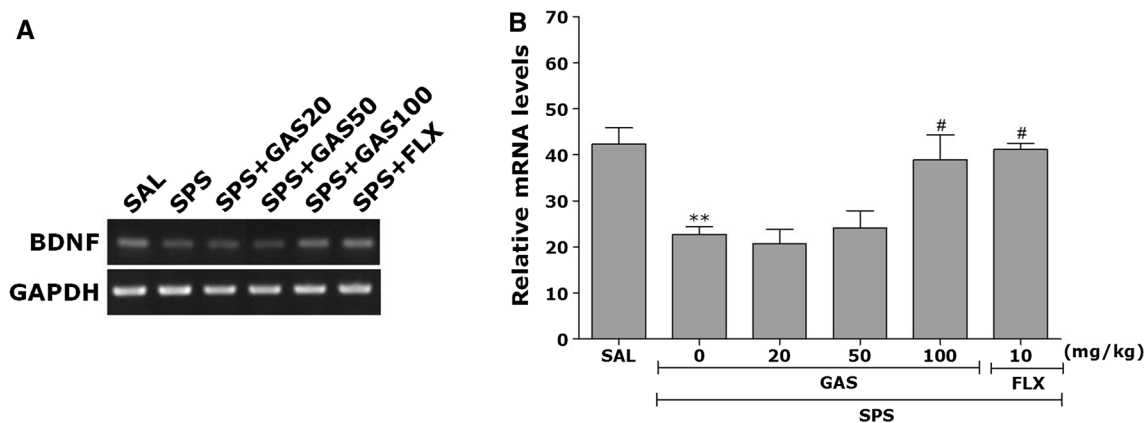


Fig. 6 Effect of GAS administration on the expression levels of BDNF mRNA in rats exposed to SPS for 14 consecutive days. PCR bands on an agarose gel and their relative intensities. ** $p < 0.01$ vs. SAL group; # $p < 0.05$ vs. SPS group

were based on the modulation of NPY in the hypothalamus and TH activity in the LC, which is the primary projection area that underlies the development of major depressive disorders via the prevention of reductions in brain levels of NE [30]. Thus, the present findings support the possibility that GAS exerts antidepressant effects in a rat model of SPS-induced PTSD.

Following the SPS procedure, the rats exhibited a gradual decrease in body weight gain and increases in serum CORT concentrations relative to the levels present immediately before testing [31, 32]; this indicates that the SPS was sufficiently stressful. Some studies suggested that exposure to SPS significantly increased plasma CORT at 14 days [33]. PTSD is associated with long-term dysregulation of the HPA axis and abnormal cortisol levels that increased in some studies of PTSD patients and decreased in others [33]. In the present study, at 14 days after SPS exposure, increased depression-like behavior and plasma CORT, and induced long-term changes in CORT occurred, consistent with other previous studies [33]. The sustained elevation in CORT levels also supports the development of depression-like behaviors, because a similar mechanism may underlie the progression or exacerbation of PTSD in humans [26]. The antidepressant effects of GAS suggest that this pharmacological compound inhibits HPA axis-associated psychological dysfunction by attenuating serum CORT levels which, in turn, would likely normalize several behavioral and neurochemical responses.

SPS was utilized to induce traumatic stress in the present study because it is one of the most reliable animal models of PTSD [26], is characterized by high face validity [26], and produces the core etiological factors of PTSD including a reduced sensorimotor gating of the startle response, depression-like behaviors, and enhanced negative feedback of the HPA axis [1]. The model of SPS used in the present study clearly increased depression-like behaviors compared

to unstressed control rats (normal rats) who also more successfully managed the stressful situations; this is consistent with previous studies [26].

The present findings are also in agreement with those of previous studies showing that SPS increases immobility during the FST [34]. In the present study, the administration of GAS significantly decreased immobility during the FST. However, there was no effect on swimming, which confirms previous results showing that GAS exerts antidepressant-like activities that do not cause deficits in motor function [26]. The FST is a valuable and reliable behavioral research model of depression in rodents and is also an important tool with which to study the neurobiological mechanisms involved in antidepressant responses [35]. The observed immobility behavior seen in the FST is similar to a state of depressed mood or helplessness and may be considered analogous to depression in humans [35]. Therefore, the present data support the possibility that GAS may have antidepressant effects in clinical populations with PTSD. Taken together, the present behavioral findings indicated that treatment with GAS exerted antidepressant-like effects in rats exposed to SPS.

Some studies suggest that the hippocampus play an important role in glucocorticoid negative feedback. Abnormalities in negative feedback are found in depression and PTSD, CORT, and NE might be involved in the pathophysiology of these disorders [36]. Animal experiments demonstrated that chronic elevation of plasma CORT caused by stress or CORT application damaged hippocampal neurons [36], and structural neuroimaging studies have shown reduced volume of the hippocampal formation in PTSD patients. Therefore, hippocampus neurons, which are known to play important roles in PTSD-like symptoms, are particularly vulnerable to PTSD-specific neuroendocrine abnormality [36]. Many studies have suggested that PTSD patients have elevated NE levels

in the brain and that this increase is correlated with severity of the disease [37]. Exaggerated central noradrenergic activity is primarily observed in response to a variety of trauma-associated stimuli rather than under resting conditions in PTSD patients [38]. This enhanced activity is likely involved in the hyperarousal of PTSD patients and their re-experience of symptom clusters [8, 39]. The present authors suggest that the SPS-induced impairments in FST performance observed in the present study were caused by enhanced noradrenergic activity in the brain. GAS appeared to inhibit the increases in hippocampal NE that can be attributed to the SPS procedure. Also, the dose of FLX (10 mg/kg, i.p.) as the positive control drug has been previously used in our laboratory and produced the increased NE levels in the hippocampus [40]. Our data suggest that the antidepressant efficacy of SSRIs such as fluoxetine may involve ability of the drug to enhance the activation of central adrenergic system by environmental stimuli [41]. Therefore, FLX attenuated behaviors and neurochemical responses associated with depression by modulating the HPA axis and the central adrenergic system, suggesting that administration of FLX, may indirectly alter catecholamine synthesis in the brain to produce physiological effects. Thus, our results indicated that FLX acts by stimulating NE synthesis in the rat brain, suggesting that an overactive central adrenergic system may contribute to depressive symptomology, and that the therapeutic action of antidepressants reverses this activity by increasing TH expression in the LC.

Additionally, because SPS-induced physiological and behavioral changes have been linked to TH expression [8], TH activity was measured in the LC, and a significant increase in TH-like immunoreactivity was observed. The TH enzyme is an important player in stress-induced activation in the central nervous system and in stress-related psychopathological conditions, including depression [26]. Therefore, GAS may regulate the manifestation of antidepressant-like behaviors in the FST via decreased expression of TH in the LC, which would be consistent with previous findings [11, 26].

A number of studies have demonstrated that the release of NPY facilitates the containment of negative consequences following exposure to stress and may increase depression-related behaviors [42]. It is thought that the NPY system plays a critical role in resilience to or recovery from stress, and that it may be protective against the development of stress-triggered neuropsychiatric disorders including PTSD [43]. In fact, the manipulation of NPY levels changes the behavioral and neurobiological responses to SPS [44]. There is decreased expression of NPY in rats after exposure to SPS, indicating that stress-induced NPY expression in the PVN activates the HPA axis in rats [4, 45]. In the present study, GAS attenuated the decrease

in NPY immunoreactivity in the PVN following exposure to SPS.

The present study also found that SPS decreased the expression of BDNF mRNA in the rat hippocampus. GAS restored hippocampal levels of BDNF mRNA, which suggests that modulation of BDNF signaling plays a role in the antidepressant actions of GAS. Meanwhile, BDNF, under both in vitro and in vivo conditions, induces increases in NPY mRNA and peptide contents [46] and regulates the function and morphology of NPY neurons [47].

GAS produced significant antidepressant-like effects closely related to the hippocampal NE levels and BDNF expression, which were similar to the classic antidepressant SSRIs. The long-lasting exaggeration of HPA activity, the persistent disruption of brain neurotransmitters, and the decrease in BDNF were all detrimental to health, driving the depression pathogenesis [48]. In this context, the neurotrophic hypothesis of major depressive disorder is accompanied by the monoamine theory [49], which postulates that high levels of monoamines are involved in the incidence of depression, particularly NE [50]. Therefore, GAS administration attenuated SPS-induced increases in hippocampal NE levels, and decreases in hippocampal mRNA expression of BDNF. Therefore, findings suggest that GAS has antidepressant-like effects on both behavioral and biochemical symptoms similar to those observed in patients with PTSD.

Therefore, our studies demonstrated that elevated levels of NE and TH in PTSD patients were exhibiting reduced expression of NPY, which is correlated with the progression or exacerbation of PTSD [8]. This suggests that GAS, like FLX, indirectly alters catecholamine synthesis in the brain [51]. The present findings indicate that GAS attenuates the behaviors and neurochemical responses associated with depression via the modulation of NPY expression and activity in the central noradrenergic system. Also, the activation of the HPA axis during stress has been shown to interact with the central noradrenergic system [26]. Therefore, the present study demonstrated that GAS suppressed the central noradrenergic system and attenuated activity in the HPA axis. Also, the co-activation of NPY and BDNF expression might influence depression-like processes, which suggests that therapies targeting PTSD-associated symptoms should target these factors [44]. Therefore, GAS may be induced behavioral, psychogenic and neurochemical changes and elucidate the depression-like behaviors induced by HPA axis dysregulation and the neurochemical interactions between NPY in the hypothalamus and BDNF in the hippocampus.

Meanwhile, PTSD also alters catecholamine release in brain areas, such as the hippocampus, hypothalamus and LC, associated with behavior of the stress response [52]. It

has been shown to be stress-induced NE release in the hippocampus as a potential risk factor for PTSD. Both PTSD also decreased hypothalamic NPY and hippocampal BDNF, whereas it increased TH in the LC. In addition to these biochemical effects, PTSD causes enduring behavioral changes, including depression-like behavior on the FST [52]. This altered physiological and behavioral response to stress may create a state of vulnerability to chronic stressors later in life and thus increase the risk for PTSD. Therefore, hippocampal neurons, hypothalamus, and LC, which are known to play an important role in PTSD-like symptoms, are particularly vulnerable to neuronal alternation induced by the SPS procedure.

In summary, the present study demonstrated that the administration of GAS was associated with antidepressant-like effects in the FST in a rat model of SPS-induced PTSD that were possibly mediated via modification of the central NE and NPY systems. Thus, GAS may be a useful alternative pharmacotherapy for the treatment of several psychiatric diseases, including PTSD.

Acknowledgments This research was supported by a Grant from the National Research Foundation of Korea funded by the Korean government (MEST) (2013R1A1A2063051).

References

1. Serova LI, Laukova M, Alaluf LG, Sabban EL (2013) Intranasal infusion of melanocortin receptor four (MC4R) antagonist to rats ameliorates development of depression and anxiety related symptoms induced by single prolonged stress. *Behav Brain Res* 250:139–147
2. Brunello N, Davidson JR, Deahl M, Kessler RC, Mendlewicz J, Racagni G, Shalev AY, Zohar J (2001) Posttraumatic stress disorder: diagnosis and epidemiology, comorbidity and social consequences, biology and treatment. *Neuropsychobiology* 43:150–162
3. Nemeroff CB, Bremner JD, Foa EB, Mayberg HS, North CS, Stein MB (2006) Posttraumatic stress disorder: a state-of-the-science review. *J Psychiatr Res* 40:1–21
4. Serova LI, Laukova M, Alaluf LG, Pucillo L, Sabban EL (2014) Intranasal neuropeptide Y reverses anxiety and depressive-like behavior impaired by single prolonged stress PTSD model. *Eur Neuropsychopharmacol* 24:142–147
5. George SA, Knox D, Curtis AL, Aldridge JW, Valentino RJ, Liberzon I (2013) Altered locus coeruleus-norepinephrine function following single prolonged stress. *Eur J Neurosci* 37:901–909
6. George SA, Stout SA, Tan M, Knox D, Liberzon I (2013) Early handling attenuates enhancement of glucocorticoid receptors in the prefrontal cortex in an animal model of post-traumatic stress disorder. *Biol Mood Anxiety Disord* 3:22–23
7. Shea A, Walsh C, Macmillan H, Steiner M (2005) Child maltreatment and HPA axis dysregulation: relationship to major depressive disorder and post traumatic stress disorder in females. *Psychoneuroendocrinology* 30:162–178
8. Sabban EL, Laukova M, Alaluf LG, Olsson E, Serova LI (2015) Locus coeruleus response to single-prolonged stress and early intervention with intranasal neuropeptide Y. *J Neurochem* 135:975–986
9. Wilson CB, McLaughlin LD, Ebenezer PJ, Nair AR, Dange R, Harre JG, Shaak TL, Diamond DM, Francis J (2014) Differential effects of sertraline in a predator exposure animal model of post-traumatic stress disorder. *Front Behav Neurosci* 8:256–267
10. Benarroch LC, Rodriguez A, Miledi R, Garcia-Alcocer G (2012) Serotonin receptors in hippocampus. *Sci World J* 2012:823493
11. Wilson CB, Ebenezer PJ, McLaughlin LD, Francis J (2014) Predator exposure/psychosocial stress animal model of post-traumatic stress disorder modulates neurotransmitters in the rat hippocampus and prefrontal cortex. *PLoS One* 9:e89104
12. Peng Z, Wang H, Zhang R, Chen Y, Xue F, Nie H, Chen Y, Wu D, Wang Y, Wang H, Tan Q (2013) Gastrodin ameliorates anxiety-like behaviors and inhibits IL-1 β level and p38 MAPK phosphorylation of hippocampus in the rat model of posttraumatic stress disorder. *Physiol Res* 62:537–545
13. Stein MB, Kline NA, Matloff JL (2002) Adjunctive olanzapine for SSRI-resistant combat-related PTSD: a double-blind, placebo-controlled study. *Am J Psychiatry* 159:1777–1779
14. Zohar J, Amital D, Miodownik C, Kotler M, Bleich A, Lane RM, Austin C (2002) Double-blind placebo-controlled pilot study of sertraline in military veterans with posttraumatic stress disorder. *J Clin Psychopharmacol* 22:190–195
15. Iwamoto Y, Morinobu S, Takahashi T, Yamawaki S (2007) Single prolonged stress increases contextual freezing and the expression of glycine transporter 1 and vesicle-associated membrane protein 2 mRNA in the hippocampus of rats. *Prog Neuropsychopharmacol Biol Psychiatry* 31:642–651
16. Wang X, Yan S, Wang A, Li Y, Zhang F (2014) Gastrodin ameliorates memory deficits in 3,3'-iminodipropionitrile-induced rats: possible involvement of dopaminergic system. *Neurochem Res* 39:1458–1466
17. Sun W, Miao B, Wang XC, Duan JH, Ye X, Han WJ, Wang WT, Luo C, Hu SJ (2012) Gastrodin inhibits allodynia and hyperalgesia in painful diabetic neuropathy rats by decreasing excitability of nociceptive primary sensory neurons. *PLoS One* 7:e39647
18. Wang Q, Chen G, Zeng S (2008) Distribution and metabolism of gastrodin in rat brain. *J Pharm Biomed Anal* 46:399–404
19. An SJ, Park SK, Hwang IK, Choi SY, Kim SK, Kwon OS, Jung SJ, Baek NI, Lee HY, Won MH, Kang TC (2003) Gastrodin decreases immunoreactivities of gamma-aminobutyric acid shunt enzymes in the hippocampus of seizure-sensitive gerbils. *J Neurosci Res* 71:534–543
20. Dai JN, Zong Y, Zhong LM, Li YM, Zhang W, Bian LG, Ai QL, Liu YD, Sun J, Lu D (2011) Gastrodin inhibits expression of inducible NO synthase, cyclooxygenase-2 and proinflammatory cytokines in cultured LPS-stimulated microglia via MAPK pathways. *PLoS One* 6:e21891
21. Manavalan A, Ramachandran U, Sundaramurthi H, Mishra M, Sze SK, Hu JM, Feng ZW, Heese K (2012) *Gastrodia elata* Blume (tianma) mobilizes neuro-protective capacities. *Int J Biochem Mol Biol* 3:219–241
22. Zhao X, Zou Y, Xu H, Fan L, Guo H, Li X, Li G, Zhang X, Dong M (2012) Gastrodin protect primary cultured rat hippocampal neurons against amyloid-beta peptide-induced neurotoxicity via ERK1/2-Nrf2 pathway. *Brain Res* 1482:13–21
23. Zeng X, Zhang S, Zhang L, Zhang K, Zheng X (2006) A study of the neuroprotective effect of the phenolic glucoside gastrodin during cerebral ischemia in vivo and in vitro. *Planta Med* 72:1359–1365
24. Hsieh MT, Wu CR, Chen CF (1997) Gastrodin and p-hydroxybenzyl alcohol facilitate memory consolidation and retrieval, but not acquisition, on the passive avoidance task in rats. *J Ethnopharmacol* 56:45–54
25. Jung JW, Yoon BH, Oh HR, Ahn JH, Kim SY, Park SY, Ryu JH (2006) Anxiolytic-like effects of *Gastrodia elata* and its phenolic constituents in mice. *Biol Pharm Bull* 29:261–265

26. Serova LI, Tillinger A, Alaluf LG, Laukova M, Keegan K, Sabban EL (2013) Single intranasal neuropeptide Y infusion attenuates development of PTSD-like symptoms to traumatic stress in rats. *Neuroscience* 236:298–312
27. Porolt RD, Martin P, Lenègre A, Fromage S, Drieu K (1990) Effects of an extract of Ginkgo Biloba (EGB 761) on “learned helplessness” and other models of stress in rodents. *Pharmacol Biochem Behav* 36:963–971
28. Gmeiner BM, Seelos CC (1995) Measurement of phosphotyrosine phosphatase activity using the Folin-Ciocalteu phenol reaction. *Biochem Mol Biol Int* 36:943–948
29. Paxinos G, Watson C (1986) The rat brain in stereotaxic coordinates, 3rd edn. Academic Press, New York, pp 54–85
30. Serova LI, Laukova M, Alaluf LG, Sabban EL (2014) Blockage of melanocortin-4 receptors by intranasal HS014 attenuates single prolonged stress-triggered changes in several brain regions. *J Neurochem* 131:825–835
31. Juven-Wetzler A, Cohen H, Kaplan Z, Kohen A, Porat O, Zohar J (2014) Immediate ketamine treatment does not prevent post-traumatic stress responses in an animal model for PTSD. *Eur Neuropsychopharmacol* 24:469–479
32. Patki G, Li L, Allam F, Solanki N, Dao AT, Alkadhi K, Salim S (2014) Moderate treadmill exercise rescues anxiety and depression-like behavior as well as memory impairment in a rat model of posttraumatic stress disorder. *Physiol Behav* 130:47–53
33. Zhang Y, Gandhi PR, Standifer KM (2012) Increased nociceptive sensitivity and nociceptin/orphanin FQ levels in a rat model of PTSD. *Mol Pain* 8:76
34. Takahashi T, Morinobu S, Iwamoto Y, Yamawaki S (2006) Effect of paroxetine on enhanced contextual fear induced by single prolonged stress in rats. *Psychopharmacology* 189:165–173
35. Cryan JF, Holmes A (2005) The ascent of mouse: advances in modelling human depression and anxiety. *Nat Rev Drug Discov* 4:775–790
36. Kohda K, Harada K, Kato K, Hoshino A, Motohashi J, Yamaji T, Morinobu S, Matsuoka N, Kato N (2007) Glucocorticoid receptor activation is involved in producing abnormal phenotypes of single-prolonged stress rats: a putative post-traumatic stress disorder model. *Neuroscience* 148:22–33
37. Geraciotti TD Jr, Baker DG, Ekhtor NN, West SA, Hill KK, Bruce AB, Schmidt D, Rounds-Kugler B, Yehuda R, Keck PE Jr, Kasckow JW (2001) CSF norepinephrine concentrations in posttraumatic stress disorder. *Am J Psychiatry* 158:1227–1230
38. Geraciotti TD Jr, Baker DG, Kasckow JW, Strawn JR, Jeffrey Mulchahey J, Dashevsky BA, Horn PS, Ekhtor NN (2008) Effects of trauma-related audiovisual stimulation on cerebrospinal fluid norepinephrine and corticotropin-releasing hormone concentrations in post-traumatic stress disorder. *Psychoneuroendocrinology* 33:416–424
39. O'Donnell T, Hegadoren KM, Coupland NC (2004) Noradrenergic mechanisms in the pathophysiology of post-traumatic stress disorder. *Neuropsychobiology* 50:273–283
40. Zhang W, Perry KW, Wong DT, Potts BD, Bao J, Tollefson GD, Bymaster FP (2000) Synergistic effects of olanzapine and other antipsychotic agents in combination with fluoxetine on norepinephrine and dopamine release in rat prefrontal cortex. *Neuropsychopharmacology* 23:250–262
41. Tianzhu Z, Shihai Y, Juan D (2014) Antidepressant-like effects of cordycepin in a mice model of chronic unpredictable mild stress. *Evid Based Complement Alternat Med* 2014:438506
42. Heilig M (2004) The NPY system in stress, anxiety and depression. *Neuropeptides* 38:213–224 **Review**
43. Wu G, Feder A, Wegener G, Bailey C, Saxena S, Charney D, Mathé AA (2011) Central functions of neuropeptide Y in mood and anxiety disorders. *Expert Opin Ther Targets* 15:1317–1331
44. Cohen H, Liu T, Kozlovsky N, Kaplan Z, Zohar J, Mathé AA (2012) The neuropeptide Y (NPY)-ergic system is associated with behavioral resilience to stress exposure in an animal model of post-traumatic stress disorder. *Neuropsychopharmacology* 37:350–363
45. Cui H, Sakamoto H, Higashi S, Kawata M (2008) Effects of single-prolonged stress on neurons and their afferent inputs in the amygdala. *Neuroscience* 152:703–712
46. Ji MH, Jia M, Zhang MQ, Liu WX, Xie ZC, Wang ZY, Yang JJ (2014) Dexmedetomidine alleviates anxiety-like behaviors and cognitive impairments in a rat model of post-traumatic stress disorder. *Behav Brain Funct* 10:28–37
47. Barnea A, Roberts J (2010) Induction of functional and morphological expression of neuropeptide Y (NPY) in cortical cultures by brain-derived neurotrophic factor (BDNF): evidence for a requirement for extracellular-regulated kinase (ERK)-dependent and ERK-independent mechanisms. *Brain Res* 919:57–69
48. Tao W, Wang H, Su Q, Chen Y, Xue W, Xia B, Duan J, Chen G (2016) Paeonol attenuates lipopolysaccharide-induced depressive-like behavior in mice. *Psychiatry Res* 238:116–121
49. Maletic V, Robinson M, Oakes T, Iyengar S, Ball SG, Russell J (2007) Neurobiology of depression: an integrated view of key findings. *Int J Clin Pract* 61:2030–2040
50. Yan WJ, Tan YC, Xu JC, Tang XP, Zhang C, Zhang PB, Ren ZQ (2015) Protective effects of silibinin and its possible mechanism of action in mice exposed to chronic unpredictable mild stress. *Biomol Ther* 23:245–250
51. Aykaç A, Aydın B, Cabadak H, Gören MZ (2012) The change in muscarinic receptor subtypes in different brain regions of rats treated with fluoxetine or propranolol in a model of post-traumatic stress disorder. *Behav Brain Res* 232:124–129
52. Bingham BC, Sheela Rani CS, Frazer A, Strong R, Morilak DA (2013) Exogenous prenatal corticosterone exposure mimics the effects of prenatal stress on adult brain stress response systems and fear extinction behavior. *Psychoneuroendocrinology* 38:2746–2757